Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, suspension

Species Used Human, K562, chronic myeloid leukemia.

Molecules Electroporated DNA: linearized (5kB) n-rasCAT plasmids

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Before the Pulse

Cell Growth Medium RPMI + 20% Fetal Calf Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest log phase

Pre-pulse Incubation None

Wash Solution RPMI

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature Room temperature

Electroporation Medium* RPMI

Cell Density 10 (7) cells / cuvette

Volume of Cells 850 µl

DNA Concentration 50 µg DNA / cuvette

DNA Resuspension Buffer Not given

Volume of DNA 50 to 300 µl

Cuvette Gap 0.4 cm

Voltage 2.0 kV

Field Strength 5 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) Ω none. NOT

Time Constant 0.4 msec

After the Pulse

Outgrowth Medium RPMI + 20% Fetal Calf Serum

Outgrowth Temperature 37 °C

Length of Incubation 2 days

Selection Method or Assay Used G418

Electroporation Efficiency Not given

Per Cent Survival Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

It is NOT RECOMMENDED to use high voltage without the Pulse Controller.

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Survey Number

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